

Design-oriented stress–strain model for FRP-confined concrete

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Abstract

External confinement by the wrapping of FRP sheets (or FRP jacketing) provides a very effective method for the retrofit of reinforced concrete (RC) columns subject to either static or seismic loads. For the reliable and cost-effective design of FRP jackets, an accurate stress–strain model is required for FRP-confined concrete. In this paper, a new design-oriented stress–strain model is proposed for concrete confined by FRP wraps with fibres only or predominantly in the hoop direction based on a careful interpretation of existing test data and observations. This model is simple, so it is suitable for direct use in design, but in the meantime, it captures all the main characteristics of the stress–strain behavior of concrete confined by different types of FRP. In addition, for unconfined concrete, this model reduces directly to idealized stress–strain curves in existing design codes. In the development of this model, a number of important issues including the actual hoop strains in FRP jackets at rupture, the sufficiency of FRP confinement for a significant strength enhancement, and the effect of jacket stiffness on the ultimate axial strain, were all carefully examined and appropriately resolved. The predictions of the model are shown to agree well with test data.

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1. Introduction

Fibre reinforced polymer (FRP) composites have found increasingly wide applications in civil engineering due to their high strength-to-weight ratio and high corrosion resistance. One important application of FRP-composites is as a confining material for concrete, in both the retrofit of existing reinforced concrete (RC) columns by the provision of an FRP jacket, and in concrete-filled FRP tubes in new construction. As a result of FRP confinement, both the compressive strength and ultimate strain of concrete can be greatly enhanced. In both types of applications, an accurate axial stress–axial strain model (referred to simply as stress–strain model hereafter) is required for FRP-confined concrete for reliable and cost-effective designs.

In early studies of FRP retrofit of RC columns, the stress–strain model of Mander et al. for steel-confined concrete [1] was directly used in the analysis of FRP-confined concrete columns [2,3]. Subsequent studies, however, showed that this direct use is inappropriate. This is because in Mander et al.'s model [1], a constant

confining pressure is assumed, which is the case for steel-confined concrete when the steel is in plastic flow, but not the case for FRP-confined concrete. As FRP composites remain linear elastic until final rupture, the lateral confining pressure in FRP-confined concrete increases continuously with the applied load.

Many investigations have been conducted into the behavior of FRP-confined concrete and as a result, a number of stress–strain models have been proposed. These models can be classified into two categories: (a) design-oriented models [4–11], and (b) analysis-oriented models [12–14]. In the first category, the compressive strength, ultimate axial strain (hereafter, often referred to as ultimate strain for brevity) and stress–strain behavior of FRP-confined concrete are predicted using closed-form equations based directly on the interpretation of experimental results. In the second category, stress–strain curves of FRP-confined concrete are generated using an incremental numerical procedure. In this second approach, an active confinement model for concrete is used to evaluate the axial stress and strain of passively confined concrete at a given confining pressure and the interaction between the concrete and the confining material is explicitly accounted for by equilibrium and radial displacement compatibility considerations. In

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the three studies cited above [12–14], the model of Mander et al. [1] was used as the active confinement model.

Although analysis-oriented models have advantages in accounting for the interaction between concrete and confining materials including both steel and FRP composites, the complexity of the incremental process prevents analysis-oriented models from direct use in design. They are, however, suitable for incorporation in computer-based numerical analysis such as non-linear finite element analysis. Compared to analysis-oriented models, design-oriented models are particularly suitable for direct application in design calculations. A simple and accurate design-oriented stress–strain model offers an approach that is familiar to engineers for determining the strength and ductility of FRP-confined RC structural members. It may be worth noting that in Eurocode 2 [15], although a stress–strain model for concrete in uniaxial compression is provided for structural analysis, a simpler idealized model is recommended for design use, which thus represents a similar differentiation between analysis-oriented and design-oriented models.

This paper is concerned with the development of a new design-oriented stress–strain model for concrete confined by an FRP jacket in which the reinforcing fibres are only or predominantly oriented in the hoop direction, so that the jacket has little longitudinal stiffness. That is, the jacket can be simplified as a unidirectional material providing only hoop resistance to any expansion of the concrete. The ultimate condition of such FRP-confined concrete is reached when the FRP ruptures due to hoop tensile stresses; failure of insufficient vertical lap joints is excluded from consideration here. Concrete-filled FRP tubes for new construction also make use of FRP confinement but they differ significantly in behavior as a result of the substantial longitudinal stiffness possessed by the tube (e.g. part of the hoop strain of the tube comes from its own Poisson's effect). Concrete confined by such FRP tubes is thus excluded from consideration here.

The paper begins with a review of the fundamental behavior of FRP-confined concrete as established by existing test results, followed by a discussion of the deficiencies of existing design-oriented stress–strain models based on available test observations. A new design-oriented stress–strain model, which overcomes these deficiencies is then presented and compared with test data.

2. Confining action of FRP jacket

In most applications, the lateral confinement provided by an FRP jacket to concrete is passive in nature. When the concrete is subject to axial compression, it expands laterally. This expansion is confined by the FRP jacket, which is loaded in tension in the hoop direction. Differ-

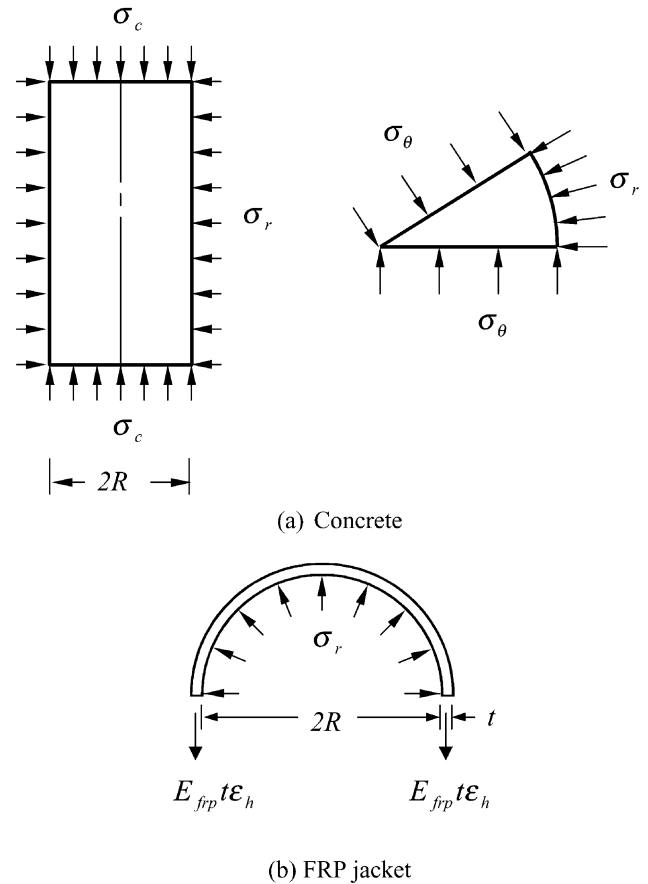


Fig. 1. Confining action of (b) FRP jacket to (a) Concrete.

ent from steel-confined concrete in which the lateral confining pressure is constant following the yielding of steel, the confining pressure provided by the FRP jacket increases with the lateral strain of concrete because FRP does not yield. The confining action in FRP-confined concrete can be schematically illustrated in Fig. 1, where all stresses are shown in their positive directions. In the concrete, compressive stresses and strains are defined as positive but in the FRP, tensile stresses and strains are positive. The lateral (radial) confining pressure acting on the concrete core σ_r is given by

$$\sigma_r = \frac{\sigma_h t}{R} = \frac{2\sigma_h t}{d} \quad (1)$$

where σ_h = tensile stress in the FRP jacket in the hoop direction, t = total thickness of the FRP jacket, and R and d = radius and diameter of the confined concrete core, respectively. If the FRP is loaded in hoop tension only, then the hoop stress in the FRP jacket σ_h is proportional to the hoop strain ϵ_h due to the linearity of FRP and is given by

$$\sigma_h = E_{frp} \epsilon_h \quad (2)$$

where E_{frp} = elastic modulus of FRP.

3. Experimental behavior of FRP-confined concrete

3.1. Test database

Many tests have been conducted on FRP-confined concrete. In the present study, a database containing the test results of 76 FRP-wrapped plain concrete circular specimens was assembled from an extensive survey of the open literature (Table 1). These 76 specimens were reported by Picher et al. [16], Watanable et al. [17], Matthys et al. [18], Purba and Mufti [19], Kshirsagar et al. [20], Rochette and Labossiere [21], Xiao and Wu [11], Aire et al. [22], Dias da Silva and Santos [23], Micelli et al. [24], Pessiki et al. [25], Wang and Cheong [26], De Lorenzis et al. [27], and Shehata et al. [28]. The specimens included in the database have diameters d from 100 mm to 200 mm and unconfined concrete strengths f'_{co} from 26.2 to 55.2 MPa. Two specimens with a very small diameter ($d=55$ mm) tested by De Lorenzis et al. [27] have been excluded. As this study is limited to normal strength concrete, 10 specimens tested by Aire et al. [22] with $f'_{co}=69$ MPa have also been excluded. It should be noted that the term 'specimen' is used loosely here for convenience, as some of the specimens represent the average performance of up to three nominally identical physical specimens.

Different types of FRP were used in the specimens in the database, namely carbon FRP (CFRP), aramid FRP (AFRP), and glass FRP (GFRP). The carbon fibres used include high strength and high modulus carbon fibres. In the following discussion, the FRP prepared from high strength carbon fibres is referred to simply as CFRP, but that prepared from high modulus carbon fibres is referred to as HM CFRP. The fibres employed were supplied in the form of unidirectional tow sheets (carbon fibres), or woven fabrics with the fibres (aramid and glass fibres) mainly in one direction. They were wrapped on concrete cylinders with the main fibres running in the hoop direction, so the resulting FRP jacket had an insignificant stiffness in the axial direction. A few specimens that were wrapped with FRP jackets with a significant stiffness in the axial direction have been excluded from the database. These include two specimens tested by Pessiki et al. [25] with glass fibres at 0 and $\pm 45^\circ$ from the hoop direction, and three specimens tested by Dias da Silva and Santos [23] with glass fibre woven fabrics having a fibre thickness of 0.094 mm and 0.040 mm in the circumferential and axial directions, respectively. For most specimens (Table 1a), the FRP properties were determined from flat coupon tensile tests [29] by researchers themselves. For the rest (Table 1b), the FRP properties were supplied by manufacturers.

Table 1 reports the compressive strength f'_{cc} and ultimate axial strain ε_{cu} of confined concrete, and the FRP hoop strain at rupture $\varepsilon_{h,rupt}$ for all specimens. The

axial strains are average values that were obtained either using strain gauges (up to three) at the mid height of the specimens or from relative displacement measurements of the middle region or between the two ends using linear-variable differential transformers (LVDTs) (up to two). The FRP hoop strains are also average values from strain gauges (up to four), or are taken to be the same as lateral strains deduced from measurements of LVDTs at the mid height of specimens except those reported by Pessiki et al. [25]. In Pessiki et al.'s study, an array of strain gauges was used to measure FRP hoop strains, and where possible, average values from a number of gauges of the critical regions (near locations of rupture) were reported. It should be noted that assuming the deformation of the confined concrete cylinder is truly axisymmetric, the lateral strain and the hoop strain of the FRP jacket are always equal in magnitude but opposite in sign according to the present sign convention. This is taken to be true in the present study, despite that a small amount of asymmetry is unavoidable in the deformation due to factors such as the inhomogeneity of concrete and eccentricity of loading.

3.2. Failure mode and FRP hoop rupture strain

All specimens included in Table 1 failed by the rupture of the FRP jacket due to hoop tension. This is the most common mode of failure for FRP-confined concrete, although premature failure due to the separation of the FRP at the vertical lap joint has also been reported for specimens with an insufficient lap length [19]. Specimens failing by a mode other than FRP rupture have been excluded from the present database.

In existing models for FRP-confined concrete, it is commonly assumed that the FRP ruptures when the hoop stress in the FRP jacket reaches its tensile strength from either flat coupon tests [29] or ring splitting tests [30] which is herein referred to as the FRP material tensile strength. This assumption is the basis for calculating the maximum confining pressure f_l (the confining pressure reached when the FRP ruptures) using the following equation:

$$f_l = \frac{2f_{frp}t}{d} \quad (3)$$

where f_{frp} = FRP material tensile strength in the hoop direction. The confinement ratio of an FRP-confined specimen is defined as the ratio of the maximum confining pressure to the unconfined concrete strength (f_l/f'_{co}).

However, experimental results show that in most cases, the FRP material tensile strength was not reached at the rupture of FRP in FRP-confined concrete. Table 2 provides the average ratios between the measured

Table 1
Test results of FRP-wrapped concrete specimens

No.	Source of data	d (mm)	L (mm)	f'_{co} (MPa)	ε_{co} (%)	Fiber type	t (mm)	f_{frp} (MPa)	E_{frp} (GPa)	ε_{cc} (%)	ε_{cu} (%)	$\varepsilon_{h,wrap}$ (%)	f'_{cc} (MPa)	f'_{cu} (MPa)	f_o (MPa)
(a) FRP properties from flat coupon tests by researchers															
1	Watanable et al. [17]	100	200	30.2	0.23	Carbon	0.17	2716	224.6		1.51	0.94	46.6		32.0
2	Watanable et al. [17]	100	200	30.2	0.23	Carbon	0.50	2873	224.6		3.11	0.82	87.2		35.0
3	Watanable et al. [17]	100	200	30.2	0.23	Carbon	0.67	2658	224.6		4.15	0.76	104.6		35.0
4	Watanable et al. [17]	100	200	30.2	0.23	HM carbon	0.14	1579	628.6		0.57	0.23	41.7		30.0
5	Watanable et al. [17]	100	200	30.2	0.23	HM carbon	0.28	1824	629.6		0.88	0.22	56.0		36.0
6	Watanable et al. [17]	100	200	30.2	0.23	HM carbon	0.42	1285	576.6		1.30	0.22	63.3		40.0
7	Watanable et al. [17]	100	200	30.2	0.23	Aramid	0.15	2589	97.1		1.58	2.36	39.0		30.0
8	Watanable et al. [17]	100	200	30.2	0.23	Aramid	0.29	2707	87.3		4.75	3.09	68.5		30.5
9	Watanable et al. [17]	100	200	30.2	0.23	Aramid	0.43	2667	87.3		5.55	2.65	92.1		35.0
10	Matthys et al. [18]	150	300	34.9	0.21	Carbon	0.12	2600	200.0		0.85	1.15	44.3		32.5
11	Matthys et al. [18]	150	300	34.9	0.21	Carbon	0.12	2600	200.0		0.72	1.08	42.2		32.5
12	Matthys et al. [18]	150	300	34.9	0.21	HM carbon	0.24	1100	420.0		0.40	0.19	41.3		31.0
13	Matthys et al. [18]	150	300	34.9	0.21	HM carbon	0.24	1100	420.0		0.36	0.18	40.7		
14	Kshirsagar et al. [20]	102	204	38.0	0.22	E-glass	1.42	363	19.9		1.73	1.74	57.0		38.1
15	Kshirsagar et al. [20]	102	204	39.4		E-glass	1.42	363	19.9		1.60	2.07	63.1		40.0
16	Kshirsagar et al. [20]	102	204	39.5		E-glass	1.42	363	19.9		1.79	1.89	60.4		40.0
17	Rochette and Labossiere [21]	100	200	42.0		Carbon	0.60	1265	82.7		1.65	0.89	73.5		
18	Rochette and Labossiere [21]	100	200	42.0		Carbon	0.60	1265	82.7		1.57	0.95	73.5		
19	Rochette and Labossiere [21]	100	200	42.0		Carbon	0.60	1265	82.7		1.35	0.80	67.6		
20	Rochette and Labossiere [21]	150	300	43.0		Aramid	1.27	230	13.6		1.11	1.53	47.3		
21	Rochette and Labossiere [21]	150	300	43.0		Aramid	2.56	230	13.6		1.47	1.39	58.9		
22	Rochette and Labossiere [21]	150	300	43.0		Aramid	3.86	230	13.6		1.69	1.33	71.0		
23	Rochette and Labossiere [21]	150	300	43.0		Aramid	5.21	230	13.6		1.74	1.18	74.4		
24	Xiao and Wu [11]	152	305	33.7		Carbon	0.38	1577	105.0		1.20	0.84	47.9		31.2
25	Xiao and Wu [11]	152	305	33.7		Carbon	0.38	1577	105.0		1.40	1.15	49.7		31.2
26	Xiao and Wu [11]	152	305	33.7		Carbon	0.38	1577	105.0		1.24	0.87	49.4		31.2
27	Xiao and Wu [11]	152	305	33.7		Carbon	0.76	1577	105.0		1.65	0.91	64.6		36.0
28	Xiao and Wu [11]	152	305	33.7		Carbon	0.76	1577	105.0		2.25	1.00	75.2		36.0
29	Xiao and Wu [11]	152	305	33.7		Carbon	0.76	1577	105.0		2.16	1.00	71.8		36.0
30	Xiao and Wu [11]	152	305	33.7		Carbon	1.14	1577	105.0		2.45	0.82	82.9		38.4
31	Xiao and Wu [11]	152	305	33.7		Carbon	1.14	1577	105.0		3.03	0.90	95.4		38.4
32	Xiao and Wu [11]	152	305	43.8		Carbon	0.38	1577	105.0	0.60	0.98	0.81	54.8	54.4	50.4
33	Xiao and Wu [11]	152	305	43.8		Carbon	0.38	1577	105.0	0.39	0.47	0.76	52.1	51.4	50.4
34	Xiao and Wu [11]	152	305	43.8		Carbon	0.38	1577	105.0	0.28	0.37	0.28	48.7	39.2	50.4
35	Xiao and Wu [11]	152	305	43.8		Carbon	0.76	1577	105.0		1.57	0.92	84.0		50.4
36	Xiao and Wu [11]	152	305	43.8		Carbon	0.76	1577	105.0		1.37	1.00	79.2		50.4
37	Xiao and Wu [11]	152	305	43.8		Carbon	0.76	1577	105.0		1.66	1.01	85.0		50.4
38	Xiao and Wu [11]	152	305	43.8		Carbon	1.14	1577	105.0		1.74	0.79	96.5		50.4
39	Xiao and Wu [11]	152	305	43.8		Carbon	1.14	1577	105.0		1.68	0.71	92.6		50.4
40	Xiao and Wu [11]	152	305	43.8		Carbon	1.14	1577	105.0		1.75	0.84	94.0		50.4
41	Xiao and Wu [11]	152	305	55.2		Carbon	0.38	1577	105.0	0.28	0.69	0.70	57.9	50.3	
42	Xiao and Wu [11]	152	305	55.2		Carbon	0.38	1577	105.0	0.41	0.48	0.62	62.9	53.1	
43	Xiao and Wu [11]	152	305	55.2		Carbon	0.38	1577	105.0	0.31	0.49	0.19	58.1	52.0	

44	Xiao and Wu [11]	152	305	55.2		Carbon	0.76	1577	105.0	0.39	1.21	0.74	74.6	73.1	68.6
45	Xiao and Wu [11]	152	305	55.2		Carbon	0.76	1577	105.0		0.81	0.83	77.6		68.6
46	Xiao and Wu [11]	152	305	55.2		Carbon	1.14	1577	105.0		1.43	0.76	106.5		61.2
47	Xiao and Wu [11]	152	305	55.2		Carbon	1.14	1577	105.0		1.45	0.85	108.0		61.2
48	Xiao and Wu [11]	152	305	55.2		Carbon	1.14	1577	105.0		1.18	0.70	103.3		61.2
49	De Lorenzis et al. [27]	120	240	43		Carbon	0.3	1028	91.1	0.87	1.16	0.70	58.5	55.6	
50	De Lorenzis et al. [27]	120	240	43		Carbon	0.3	1028	91.1	0.82	0.95	0.80	65.6	63.5	
51	De Lorenzis et al. [27]	150	300	38		Carbon	0.45	1028	91.1	0.71	0.95	0.80	62		
52	De Lorenzis et al. [27]	150	300	38		Carbon	0.45	1028	91.1	1.24	1.35	0.80	67.3		
(b) FRP properties from manufacturers															
53	Picher et al. [16]	152	304	39.7		Carbon	0.36	1266	83.0		1.070	0.84	56.0		42.5
54	Purba and Mufti [19]	191	788	27.1		Carbon	0.22	3483	230.5		0.576	0.67	53.9		43.8
55	Aire et al. [22]	150	300	42.0		Glass	0.149	3000	65.0	0.30	0.73	0.55	41.0	35.7	
56	Aire et al. [22]	150	300	42.0		Glass	0.447	3000	65.0		1.74	1.30	61		35.7
57	Aire et al. [22]	150	300	42.0		Glass	0.894	3000	65.0		2.5	1.10	85		41.1
58	Aire et al. [22]	150	300	42.0		Carbon	0.117	3900	240.0		1.1	0.95	46		40.7
59	Aire et al. [22]	150	300	42.0		Carbon	0.351	3900	240.0		2.26	1.05	77		40.7
60	Aire et al. [22]	150	300	42.0		Carbon	0.702	3900	240.0		3.23	1.06	108		40.7
61	Dias da Silva and Santos [23]	150	600	28.2		Carbon	0.111	3700	240.0		0.39	0.26	31.4		28.2
62	Dias da Silva and Santos [23]	150	600	28.2		Carbon	0.222	3700	240.0		2.05	1.18	57.4		26.8
63	Dias da Silva and Santos [23]	150	600	28.2		Carbon	0.333	3700	240.0		2.59	1.14	69.5		29.6
64	Dias da Silva and Santos [23]	150	600	28.2		HM carbon	0.167	3000	390.0		0.75	0.37	41.5		24
65	Dias da Silva and Santos [23]	150	600	28.2		HM carbon	0.334	3000	390.0		1.81	0.69	65.6		28.2
66	Dias da Silva and Santos [23]	150	600	28.2		HM carbon	0.501	3000	390.0		1.69	0.64	79.4		28.2
67	Micelli et al. [24]	102	204	37		Carbon	0.16	3790	227.0		1.02	1.2	60		37
68	Micelli et al. [24]	102	204	32		Glass	0.35	1520	72.0		1.25	1.25	52		33
69	Pessiki et al. [25]	152	610	26.2	0.22	E-glass	1.00	383	21.6		1.30	1.15	38.4		31.5
70	Pessiki et al. [25]	152	610	26.2	0.22	E-glass	2.00	383	21.6		1.82	1.24	52.5		31.5
71	Pessiki et al. [25]	152	610	26.2	0.22	Carbon	1.00	580	38.1		1.44	0.81	50.6		33.9
72	Pessiki et al. [25]	152	610	26.2	0.22	Carbon	2.00	580	38.1		1.65	0.72	64.0		33.9
73	Wang and Cheong [26]	200	600	27.9	0.16	Carbon	0.36	4400	235.0		1.52	0.85	82.8		32.5
74	Wang and Cheong [26]	200	600	27.9	0.16	Carbon	0.36	4400	235.0		1.43	1.07	81.2		32.5
75	Shehata et al. [28]	150	300	29.8	0.21	Carbon	0.165	3550	235.0		1.23	1.23	57.0		33
76	Shehata et al. [28]	150	300	29.8	0.21	Carbon	0.33	3550	235.0		1.74	1.19	72.1		34

Table 2
Average hoop rupture strain ratios

Type of fibre	No. of specimens	FRP material ultimate tensile strain ε_{frp} from coupon tests		Ratio of hoop rupture strain to FRP material ultimate tensile strain $\varepsilon_{h,rupt}/\varepsilon_{frp}$ (%)	
		Average	S.D.	Average	S.D.
CFRP	52	0.0148	0.0015	58.6	15.3
High modulus CFRP	8	0.0045	0.0027	78.8	16.8
AFRP	7	0.0223	0.0068	85.1	9.5
GFRP	9	0.0280	0.0136	62.4	36.4
Total	76	0.0160	0.0080	63.2	20.5

hoop strain at FRP rupture $\varepsilon_{h,rupt}$ and the FRP material ultimate tensile strain ε_{frp} for a number of categories. It is seen that the average ratio differs for specimens confined by a different type of FRP, and has a value of 0.63 when all specimens of the present database are considered together. Thus, the maximum confining pressure given by Eq. (3) is only a nominal value. The actual maximum confining pressure should be given by

$$f_{l,a} = \frac{2E_{frp}t\varepsilon_{h,rupt}}{d} \quad (4)$$

where $f_{l,a}$ is the actual maximum confining pressure. The actual confinement ratio is then given by the ratio between $f_{l,a}$ and f'_{co} .

Table 2 indicates that the assumption that the FRP ruptures when the stress in the jacket reaches the FRP material tensile strength is invalid for concrete confined by FRP wraps. Fig. 2 shows the ratio of FRP hoop rupture strain from tests to the FRP material ultimate tensile strain against the actual confinement ratio $f_{l,a}/f'_{co}$, which illustrates the great scatter displayed by the data. In particular, this figure indicates that in concrete confined by a small amount of FRP, premature rupture of the FRP at a hoop strain much lower than the material ultimate tensile strain is very likely. For example, the four specimens with very low levels of FRP confinement failed with hoop rupture strains below 20% of the material ultimate tensile strain. The maximum value of $f_{l,a}/f'_{co}$ of these four specimens is only 0.034.

The difference between the FRP tensile strength or ultimate strain from material tests and that reached in tests of FRP-confined concrete specimens has been discussed in a number of recent papers [13,25,31–33]. Several causes have been suggested for this phenomenon. The two main causes are believed to be (a) deformation localization in cracked concrete leading to a non-uniform stress distribution in the FRP jacket and thus premature rupture of FRP, and (b) the effect of

curvature of an FRP jacket on the tensile strength of FRP. Shahawy et al. [31] suggested that ring splitting tests could provide a closer estimation of the hoop rupture strain of FRP. Thorough studies on these aspects are not yet available. The hoop rupture strains reported in the existing literature are in general average values around the circumference, so strain distributions around the circumference in FRP-confined specimens are unclear at this stage.

3.3. Stress–strain response

It has been well recognized that the stress–strain curve of FRP-confined concrete features a monotonically ascending bi-linear shape as shown in Fig. 3a, if the amount of FRP exceeds a certain threshold. Such FRP-confined concrete is said to be sufficiently confined. This type of stress–strain curves (the increasing type) was observed in the vast majority of the tests covered by the present database. With this type of stress–strain curves, both the compressive strength and the ultimate strain are reached at the same point and are significantly enhanced. However, existing tests have also shown that in some cases such a bi-linear stress–strain behavior cannot be expected. Instead, the stress–strain curve features a post-peak descending branch and the compressive strength is reached before FRP rupture (the decreasing type). This decreasing type of stress–strain curves can be further differentiated in terms of the stress in concrete at the ultimate strain f'_{cu} (Fig. 3b,c). If the stress–strain curve terminates at a concrete stress f'_{cu} above the compressive strength of unconfined concrete f'_{co} as illustrated in Fig. 3b, the FRP confinement is still sufficient to lead to strength enhancement. Such concrete is also referred to as sufficiently confined concrete in the present study. However, if the stress–strain curve terminates at a stress $f'_{cu} < f'_{co}$ as illustrated in Fig. 3c, the specimen is said to be insufficiently confined, where

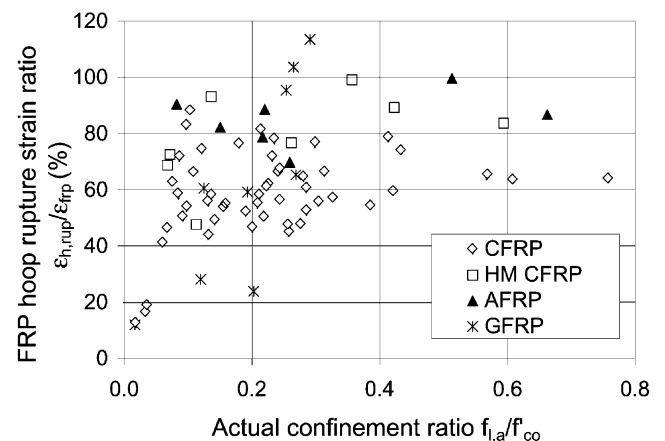


Fig. 2. Hoop strains at rupture of FRP jackets.

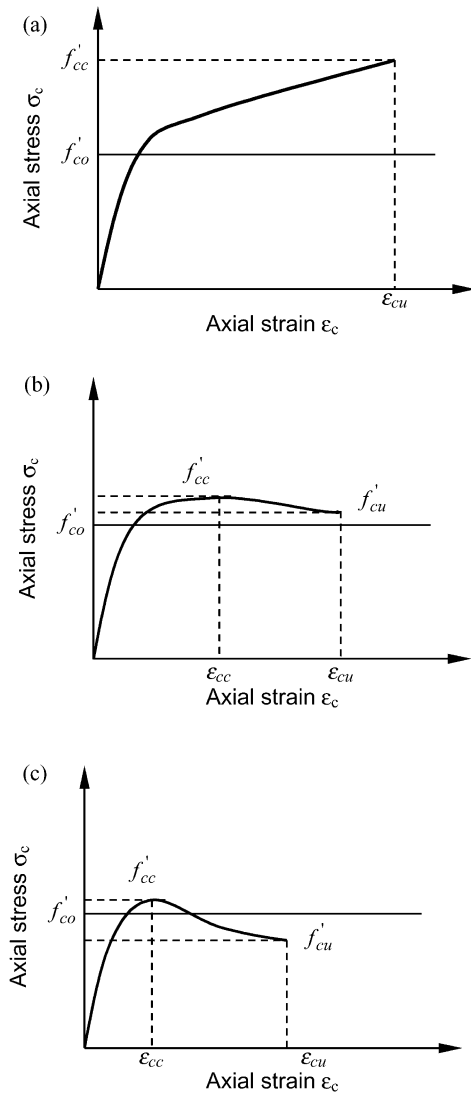


Fig. 3. Classification of stress–strain curves of FRP-confined concrete. (a) Increasing type; (b) Decreasing type with $f'_{cu} > f'_{co}$; (c) Decreasing type with $f'_{cu} < f'_{co}$.

little strength enhancement can be expected. The behavior of concrete with insufficient FRP confinement has been observed in some tests conducted by Xiao and Wu [11] and Aire et al. [22]. For those specimens with the compressive strength reached before FRP rupture, Table 1 also provides the axial strain at the peak stress ϵ_{cc} and the stress of concrete at the ultimate strain f'_{cu} where available.

A set of stress–strain curves of concrete confined by different amounts of FRP is shown in Fig. 4, using the test data obtained by Xiao and Wu [11], where the axial stress is normalized by the unconfined concrete strength. In Fig. 4, specimen A had an unconfined concrete strength of 55.2 MPa and was wrapped with one layer of CFRP (specimen 41 of Table 1a). Specimens B, C and D had an unconfined concrete strength of 43.8 MPa

and were wrapped with one, two, and three layers of CFRP, respectively, (specimens 32, 35 and 40 in Table 1a). Specimen E had an unconfined concrete strength of 33.7 MPa and was wrapped with three layers of CFRP (specimen 31 in Table 1a). The actual confinement ratios of these five specimens are 0.067, 0.097, 0.221, 0.302 and 0.421, respectively. Further details of these specimens can be found in Table 1a. Specimen A exhibits insignificant strength enhancement and the stress–strain curve terminates at a stress below the unconfined concrete strength (Fig. 4). Specimen B also has a decreasing stress–strain response after the peak stress, but the stress at the ultimate strain is higher than the unconfined concrete strength. The behavior of this specimen thus belongs to the type illustrated in Fig. 3b. Specimens C, D and E all has a stress–strain curve of the increasing type. For these three specimens, the enhancement in both the compressive strength and the ultimate strain increases with the amount of confinement.

The volumetric change of confined concrete under axial compression can be represented by the volumetric strain ϵ_v , which is defined by

$$\epsilon_v = \epsilon_c + \epsilon_r + \epsilon_\theta = \epsilon_c + 2\epsilon_r \quad (5)$$

where ϵ_θ = circumferential strain and ϵ_r = lateral (radial) strain. It is commonly known that unconfined concrete in axial compression experiences a volumetric reduction or compaction up to 90% of the peak stress, but thereafter the concrete shows volumetric expansion or dilation which becomes unstable after the peak stress [34,35]. Unstable dilation has also been observed in actively confined concrete in tri-axial compression tests [36,37]. Recently, Mirmiran and his co-authors [6,35] compared the volumetric responses of FRP-confined concrete with those of plain concrete and steel-confined concrete. They demonstrated that for steel-confined concrete, unstable dilation occurs when steel yields, but for

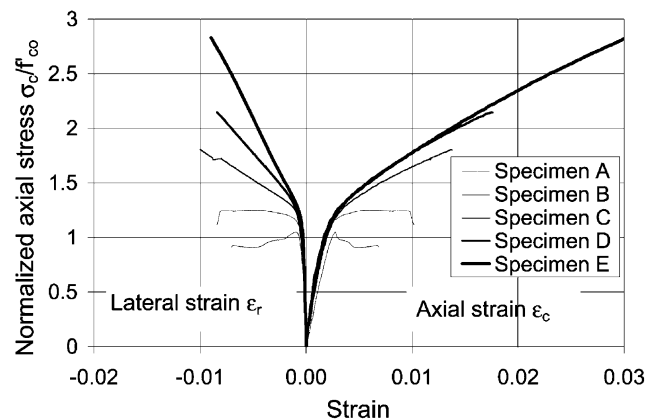


Fig. 4. Typical stress–strain curves of FRP-confined concrete.

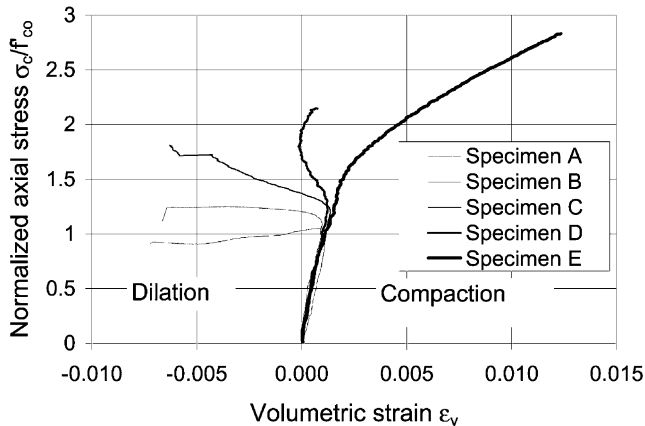


Fig. 5. Volumetric responses of FRP-confined concrete.

FRP-confined concrete, the linearly increasing hoop stress of FRP can eventually curtail the dilation if the amount of FRP is large enough.

Fig. 5 illustrates the volumetric changes of FRP-confined concrete using the test data of Xiao and Wu [11] for the five specimens shown in Fig. 4. A positive volumetric strain indicates compaction while a negative value corresponds to dilation. It can be seen that for specimens A, B and C, the volumetric strain changes from compaction to dilation at an axial stress above the compressive strength of unconfined concrete, and this dilation continues to increase until failure. For specimen D, dilation is taken over by compaction at a normalized axial stress of approximately 1.75. For specimen E, no dilation is found during the entire loading history. The different volumetric responses of FRP-confined concrete from steel-confined concrete are due to the linear elastic behavior of FRP. This linearity of FRP provides a continuously increasing confining pressure until rupture, and limits the lateral strain of confined concrete to the hoop rupture strain of FRP. As a result, concrete confined by a large amount of FRP may not show dilation at all.

3.4. Minimum amount of FRP for sufficient confinement

As explained above, FRP-confined concrete with a stress–strain curve of the decreasing type and with a concrete stress at the ultimate strain below the compressive strength of unconfined concrete is taken to be insufficiently confined, as in such cases little strength enhancement can be expected from the FRP confinement and the FRP is likely to rupture at a small hoop strain. The latter phenomenon, believed to be due to the sensitivity of a weak jacket to the non-uniform deformation of concrete, is particularly important as it means that the use of such a weak jacket leads to little strength or strain enhancement, and any enhancement cannot be reliably predicted. It is, therefore, recommended here

that such weak confinement should not be allowed in practical design. Consequently, the new stress–strain model presented in Section 5 is intended for application to sufficiently confined concrete only. For this limitation to be observed in practical design, the minimum amount of FRP deemed necessary to achieve sufficient confinement needs to be defined.

The issue of insufficient confinement has been discussed in a number of previous papers. Mirmiran et al. [38] suggested that for FRP-confined rectangular concrete specimens with rounded corners, enhancement in the compressive strength of confined concrete should not be expected if the following modified confinement ratio (MCR) is less than 0.15:

$$MCR = \left(\frac{2R_c}{D} \right) \frac{f_l}{f'_{co}} \quad (6)$$

where R_c = radius of rounded corners and D = side length of confined square section, f'_{co} = compressive strength of unconfined concrete and f_l = equivalent maximum confining pressure. For circular specimens, this criterion reduces to $f_l/f'_{co} < 0.15$. However, Spoelstra and Monti [13] showed that the stress of concrete at the ultimate strain f'_{cu} falls below f'_{co} if $f_l/f'_{co} < 0.07$ using their analysis-oriented model. Besides, based on experimental data, Xiao and Wu [11] suggested that for FRP-confined concrete with $E_{frp}t/Rf'^2_{co} < 0.2$ (MPa^{-1}), a post-peak descending branch could be expected.

In Table 1, a total of five specimens have a concrete stress at the ultimate strain f'_{cu} below the unconfined concrete strength f'_{co} . The nominal confinement ratio f_l/f'_{co} is less than 0.15 for thirteen of the specimens, including four with $f'_{cu}/f'_{co} < 1$ (Fig. 6a). Another specimen showing $f'_{cu}/f'_{co} < 1$ has a nominal confinement ratio of $f_l/f'_{co} = 0.18$. Fig. 6b shows that in terms of the actual confinement ratio, of the seven specimens with an actual confinement ratio $f_{l,a}/f'_{co} < 0.07$, five have $f'_{cu}/f'_{co} < 1$ (Fig. 6b). The maximum value of $f_{l,a}/f'_{co}$ for these five specimens is 0.069. Fig. 7 shows that the criterion of Xiao and Wu [11] for insufficient confinement ($E_{frp}t/Rf'^2_{co} < 0.2$) is met by five specimens, including four with $f'_{cu}/f'_{co} < 1$ and one with $f'_{cu}/f'_{co} = 1.01$. For the other specimen with $f'_{cu}/f'_{co} < 1.01$, $E_{frp}t/Rf'^2_{co} = 0.275$.

Based on the above discussion, it can be concluded that in judging whether sufficient confinement is available, Spoelstra and Monti's [13] criterion for insufficient confinement can be used, with the maximum confining pressure being the actual rather than the nominal value. That is, FRP-confined concrete with an actual confinement ratio $f_{l,a}/f'_{co} \geq 0.07$ can be taken to be sufficiently confined.

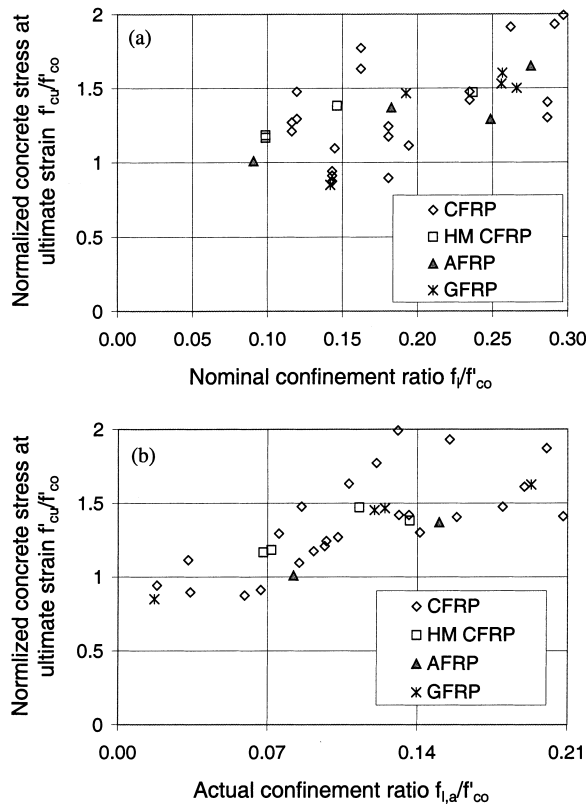


Fig. 6. Stress of concrete at ultimate strain vs. confinement ratio. (a) Variation with nominal confinement ratio; (b) Variation with actual confinement ratio.

4. Deficiencies of existing design-oriented stress–strain models

4.1. Shape of stress–strain curve

Existing design-oriented stress–strain models for FRP-confined concrete [4–11] have adopted different approximations to a typical bilinear stress–strain curve. In the models of Karbhari and Gao [5] and Xiao and Wu [11], the two portions of a bilinear curve are approximated using two straight lines. This approach is simple but not realistic. Samaan et al. [6] proposed a model for FRP-confined concrete in which the nearly linear second portion of the stress–strain curve is characterized by its slope E_2 and its intercept with the stress axis. Another salient feature of this model is that the stress–strain curve is represented by a single equation, with the transition from the first portion to the second portion being controlled by a shape parameter n . The use of a single equation, however, necessarily leads to an equation of a more complex form. Toutanji [9] and Saafi et al. [8] proposed an alternative form for the stress–strain curve, in which the two portions of a bilinear curve are approximated using two separate equations, with both equations producing a curved

shape. A smooth transition between the two portions is also provided. Based on the same general equations, two models were proposed separately for FRP-wrapped concrete and concrete-filled FRP tubes by calibrating the model parameters with corresponding test data. The models of Samaan et al. [6], Toutanji [9] and Saafi et al. [8] can predict the shape of a bilinear stress–strain curve reasonably closely, provided their predictions of the compressive strength and ultimate strain are accurate. However, the relative complexity of these three models in form means inconvenience or difficulty in section analysis for the determination of section capacity or ductility, where integration of the stress distribution over the section is required.

Miyauchi et al. [7] used Hognestad's parabola [39] followed by a straight line to describe both the increasing and decreasing types of stress–strain curves of FRP-confined concrete. This parabola, given by the following equation, is commonly adopted in codes of practice such as BS 8110 [40] and Eurocode 2 [15] to describe the ascending part of the stress–strain curve of unconfined concrete for design use:

$$\sigma_c = f'_{co} \left[\frac{2\varepsilon_c}{\varepsilon_{co}} - \left(\frac{\varepsilon_c}{\varepsilon_{co}} \right)^2 \right] \quad (7)$$

where σ_c and ε_c are the axial stress and strain, respectively, and ε_{co} is the axial strain at the peak stress of concrete. However, the direct use of Hognestad's parabola as adopted in Miyauchi et al.'s model [7] cannot reflect the process of gradual development of confinement. In fact, the FRP confinement is activated once micro-cracks in concrete are initiated under loading. Lillistone and Jolly [10] attempted to account for this effect in their stress–strain model for concrete-filled FRP tubes in which the first portion of the stress–strain curve is described using Hognestad's parabola plus an additional term related to the hoop stiffness of the FRP

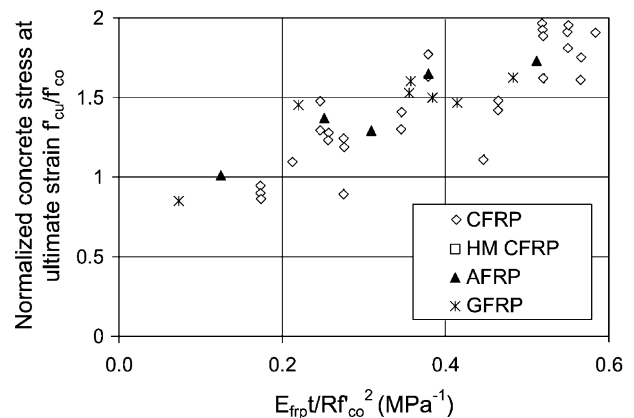


Fig. 7. Stress of concrete at ultimate strain vs. Xiao and Wu's confinement stiffness parameter.

tube ($E_{\text{frp}}t/R$), while the second portion is a straight line. This additional term used to account for the effect of confinement (not the contribution of the longitudinal stiffness), being equal to $1.282 E_{\text{frp}}t\epsilon_c/R$, means that the initial slope of the predicted stress–strain curve can be significantly greater than that of unconfined concrete, which is obviously not supported by the test results.

In summary, the present authors believe the best approach for describing a typical bilinear stress–strain curve of FRP-confined concrete is to use a modified parabola for the first portion and a straight line for the second portion, with the various parameters being dependent on the FRP properties. The modified parabola should be able to reflect the gradual development of confinement as axial stress increases. This same view has also recently been expressed by Monti [41]. Such a parabola leads to ease in section analysis where integration of the stress–strain curve is necessary and offers an approach that is familiar to engineers. For the definition of the linear second portion, the use of its slope and its intercept with the stress axis as done by Samaan et al. [6] offers a rational and simple approach.

4.2. Definition of ultimate condition

Central to any stress–strain model for FRP-confined concrete is the determination of the ultimate condition of FRP-confined concrete, which is reached when the FRP ruptures. This ultimate condition is characterized by two parameters: the ultimate axial strain and the corresponding stress level, which is generally but not always the compressive strength of FRP-confined concrete. There are three major deficiencies in existing design-oriented stress–strain models in predicting the ultimate condition of confined concrete.

Firstly, it is commonly assumed that rupture of FRP occurs when the hoop stress in the FRP jacket reaches the tensile strength determined from material tests, with the only exception being Xiao and Wu's model [11]. This assumption is, however, not valid as shown in the preceding section, and leads to difficulty in producing a unified stress–strain model for FRP-confined concrete as the ratio of hoop rupture strain to FRP material tensile strain varies with the type of FRP (Table 2). Existing analysis-oriented models also suffer from this deficiency, so they are also incapable of accurate predictions of the ultimate condition.

Secondly, the effect of the stiffness of the FRP jacket on the ultimate condition has not been well established and explicitly accounted for, although it is implied to some degree in the ultimate strain equations of Samaan et al. [6], Toutanji [9] and Saafi et al. [8]. The stiffness of the FRP jacket in fact has an important effect on the stress–strain response of FRP-confined concrete, particularly the ultimate axial strain as shown later in the paper.

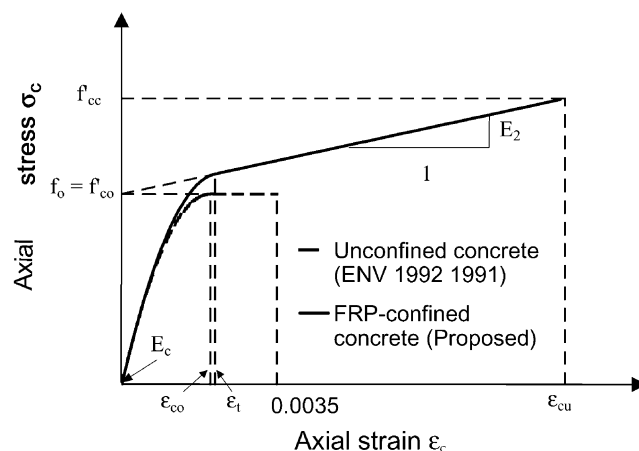


Fig. 8. Proposed stress–strain model for FRP-confined concrete.

Thirdly, as a result of the above two deficiencies and due to the use of a limited database, there is room for improvement to the accuracy of the predictive equations for the compressive strength and ultimate strain of FRP-confined concrete in existing design-oriented models using a larger test database.

In summary, improvements to existing stress–strain models in defining the ultimate condition should be made in three aspects: (a) the actual hoop rupture strain should be used instead of the ultimate material tensile strain; (b) the effect of jacket stiffness should be explicitly and properly reflected; and (c) the ultimate strain and compressive strength equations should be based on the largest test database which can be assembled from the open literature.

5. Assumptions and general equations of new model

5.1. Assumptions

A new stress–strain model is proposed here for FRP-confined concrete based on the various observations discussed in the preceding sections. The basic assumptions of this simple model are: (i) the stress–strain curve consists of a parabolic first portion and a straight-line second portion, as given in Fig. 8; (ii) the slope of the parabola at $\epsilon_c = 0$ (initial slope) is the same as the elastic modulus of unconfined concrete E_c ; (iii) the non-linear part of the first portion is affected to some degree by the presence of an FRP jacket; (iv) the parabolic first portion meets the linear second portion smoothly (i.e. there is no change in slope between the two portions where they meet); (v) the linear second portion ends at a point where both the compressive strength and the ultimate axial strain of confined concrete are reached.

These basic assumptions of the proposed model are in accordance with the test observations of FRP-confined concrete with a monotonically increasing stress–strain

curve as illustrated in Fig. 3a. The first assumption leads to a stress–strain curve which is similar to those adopted by existing design codes for unconfined concrete and thus familiar to engineers. The second assumption is to account for the fact that the initial stiffness of FRP-confined concrete is little affected by the FRP due to the passive nature of confinement. The third assumption is to reflect the fact that the FRP confinement is activated when the behavior of the concrete becomes non-linear. This third assumption makes the new model different from Miyauchi et al.'s model [7] in which the shape of the parabola remains the same as that for unconfined concrete and is not affected by the FRP confinement at all. The fourth assumption ensures a smooth stress–strain curve, while the last assumption is obviously valid for FRP-confined concrete with a monotonically increasing stress–strain curve. For FRP-confined concrete whose behavior is as illustrated in Fig. 3b, the last assumption is not reflective of reality, but the present model provides a good approximation for design use. It may be noted that even for unconfined concrete, the design stress–strain curve is represented by a parabola followed by a horizontal straight line in both BS 8110 [40] and Eurocode 2 [15], despite that test stress–strain curves display a descending post-peak branch.

Based on the assumptions listed above, the proposed stress–strain model for FRP-confined concrete is given by the following expressions:

$$\sigma_c = E_c \varepsilon_c - \frac{(E_c - E_2)^2}{4f_o} \varepsilon_c^2 \quad \text{for } 0 \leq \varepsilon_c \leq \varepsilon_t \quad (8a)$$

and

$$\sigma_c = f_o + E_2 \varepsilon_c \quad \text{for } \varepsilon_t \leq \varepsilon_c \leq \varepsilon_{cu} \quad (8b)$$

where f_o = intercept of the stress axis by the linear second portion. The parabolic first portion meets the linear second portion with a smooth transition at ε_t , which is given by

$$\varepsilon_t = \frac{2f_o}{(E_c - E_2)} \quad (9)$$

where E_2 is the slope of the linear second portion, given by

$$E_2 = \frac{f'_{cc} - f_o}{\varepsilon_{cu}} \quad (10)$$

where f'_{cc} = compressive strength of confined concrete.

This proposed model allows the use of test values or values suggested by design codes for the elastic modulus of unconfined concrete. The three parameters yet to be

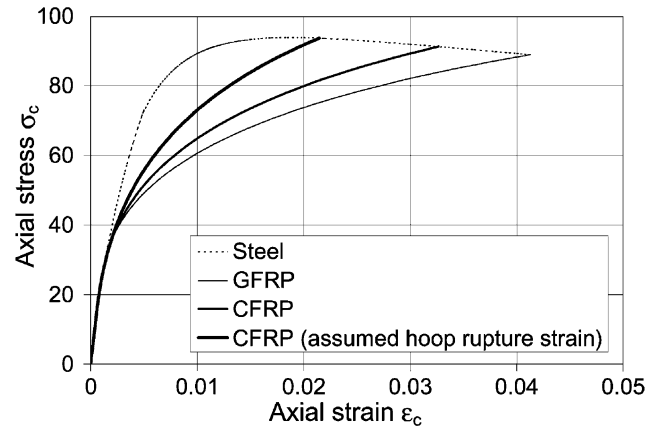


Fig. 9. Stress–strain curves predicted by Spoelstra and Monti's analysis-oriented model for concrete confined by different materials.

defined are: the ultimate strain ε_{cu} , the compressive strength f'_{cc} , and the intercept of the stress axis by the second linear portion f_o .

6. Ultimate strain of FRP-confined concrete

6.1. Theoretical basis

Studies on actively confined concrete and steel-confined concrete showed that the axial strain at the compressive strength of confined concrete ε_{cc} can be linearly related to the maximum confining pressure [42,43]. This approach has been adopted in some design-oriented models [5,7] for predicting the axial strain at the compressive strength of FRP-confined concrete ε_{cc} , which in most cases is also the ultimate strain of FRP-confined concrete ε_{cu} . A deficiency of this approach is that the stiffness of the confining jacket is not properly accounted for. This seems not important for steel-confined concrete because the elastic moduli of all types of steel are similar, but is important for FRP-confined concrete as the elastic modulus of FRP varies over a wide range. A number of studies [6,13,44] have noticed that similar levels of lateral pressure do not result in similar ultimate strains of confined concrete. Although the effect of jacket stiffness has not been properly accounted for in design-oriented models, it is always accurately represented in analysis-oriented models through equilibrium and compatibility considerations of the concrete and the jacket. This issue can thus be explained by making use of predictions from an analysis-oriented model.

Fig. 9 shows four stress–strain curves predicted by the analysis-oriented model of Spoelstra and Monti [13] for concrete cylinders confined by three different confining materials: steel, CFRP and GFRP with their properties given in Table 3. For confinement by CFRP, predictions are provided for two scenarios: the FRP

Table 3

Properties of confining materials used for predictions shown in Fig. 9

Confining material	Elastic modulus (MPa)	Rupture or yield stress (MPa)	Rupture or yield strain (%)	Thickness of confining jacket (mm)
Steel	2×10^5	300	0.15	4
CFRP	2.35×10^5	3530	1.5	0.34
CFRP (actual rupture strain)	2.35×10^5	2115	0.9	0.567
GFRP	23 100	462	2.0	2.6

ruptures at its material ultimate tensile strain from coupon tests and the FRP ruptures at an assumed hoop rupture strain of 60% of the material ultimate tensile strain. The compressive strength of unconfined concrete is 35 MPa, while the diameter of the cylinders is 150 mm. For all four cylinders, the FRP jackets are assumed to supply the same ultimate tensile capacity in the hoop direction and thus the same maximum confining pressure, but they have different stiffnesses. The substantial differences between the predicted responses including the ultimate strain are due to the differences in the stiffness of the four jackets only. While one may argue that analysis-oriented models can be inaccurate, the differences as shown in Fig. 9 are obviously too large to be attributed to the inaccuracy of the active confinement model since the same concrete is modeled and the stress–strain curve predicted by Spoelstra and Monti's model [13] for a similar CFRP-wrapped specimen has previously been shown to match the test curve closely [33].

The dependence of the ultimate strain of FRP-confined concrete on the stiffness of the confining jacket can also be shown by examining the constitutive model for concrete under a triaxial state of stress proposed by Ottosen [45]. This model is based on non-linear elasticity, with the properties of concrete being represented by the secant values of elastic modulus and Poisson's ratio. This is the constitutive model recommended by the CEB-FIP Model Code 1990 [46] for concrete under multi-axial stresses. The basic equations of the model are:

$$\varepsilon_1 = \frac{1}{E_{\text{sec}}} [\sigma_1 - \nu_{\text{sec}}(\sigma_2 + \sigma_3)] \quad (11a)$$

$$\varepsilon_2 = \frac{1}{E_{\text{sec}}} [\sigma_2 - \nu_{\text{sec}}(\sigma_1 + \sigma_3)] \quad (11b)$$

$$\varepsilon_3 = \frac{1}{E_{\text{sec}}} [\sigma_3 - \nu_{\text{sec}}(\sigma_1 + \sigma_2)] \quad (11c)$$

where E_{sec} =secant modulus of elasticity and ν_{sec} =secant Poisson's ratio. For confined concrete, $\varepsilon_1 = \varepsilon_c =$

axial strain of concrete, $\varepsilon_2 = \varepsilon_r$ =lateral (radial) strain of concrete $= \varepsilon_3 = \varepsilon_\theta$ =circumferential strain of concrete, $\sigma_1 = \sigma_c$ =compressive stress of concrete, and $\sigma_2 = \sigma_3 = \sigma_r$ =lateral confining pressure. The following equation can then be obtained from Eqs. (11a), (11b) and (11c):

$$\varepsilon_c = -\frac{\varepsilon_r}{\nu_{\text{sec}}} + \frac{(1 - \nu_{\text{sec}} - 2\nu_{\text{sec}}^2)\sigma_r}{\nu_{\text{sec}}E_{\text{sec}}} \quad (12)$$

Further, as σ_r can be expressed as a function of the hoop strain in the FRP according to Eq. (1) and Eq. (2), there is

$$\varepsilon_c = -\frac{\varepsilon_r}{\nu_{\text{sec}}} + \frac{(1 - \nu_{\text{sec}} - 2\nu_{\text{sec}}^2)E_{\text{frp}}t\varepsilon_h}{\nu_{\text{sec}}E_{\text{sec}}R} \quad (13)$$

Under the ultimate condition of FRP rupture, $\varepsilon_r = -\varepsilon_{h,\text{rup}}$. The secant modulus of elasticity of concrete under the ultimate condition E_{secu} is given by [45].

$$E_{\text{secu}} = \frac{E_{\text{seco}}}{1 + 4(A - 1)x} \quad (14)$$

where E_{seco} =secant modulus of elasticity at the compressive strength of unconfined concrete and $=f'_{co}/\varepsilon_{co}$; $A=2$ if the Hognestad's [39] parabola is assumed for the ascending part of the stress–strain curve of unconfined concrete; and x is given by

$$x = (\sqrt{J_2}/f'_{co})_f - 1/\sqrt{3} \quad (15)$$

where J_2 is the second invariant of the deviatoric stresses and $=\frac{1}{3} \times (\sigma_c - \sigma_r)^2$ for confined concrete. The term $(\sqrt{J_2}/f'_{co})_f$ denotes the value of the invariant under the ultimate condition of FRP-confined concrete, so $\sigma_c = f'_{cc}$ and $\sigma_r = f_{l,a}$. The compressive strength of confined concrete can be expressed in the following common form [47]:

$$\frac{f'_{cc}}{f'_{co}} = 1 + k_1 \frac{f_{l,a}}{f'_{co}} \quad (16)$$

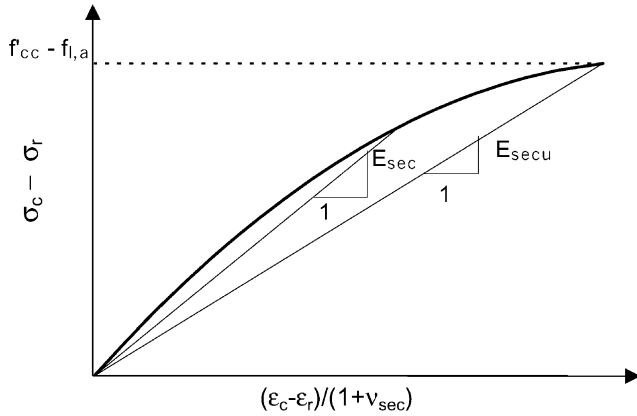


Fig. 10. Definition of secant modulus of elasticity.

where k_1 = confinement effectiveness coefficient. Eq. (15) then becomes

$$x = \frac{(k_1 - 1)}{\sqrt{3}} \left(\frac{f_{l,a}}{f'_{co}} \right) = \frac{(k_1 - 1)}{\sqrt{3}} \left(\frac{E_{frp} t}{E_{seco} R} \right) \left(\frac{\epsilon_{h,rup}}{\epsilon_{co}} \right) \quad (17)$$

Substituting Eqs. (14) and (17) into Eq. (13), the following equation can be obtained for the normalized ultimate strain of confined concrete:

$$\begin{aligned} \frac{\epsilon_{cu}}{\epsilon_{co}} = & \frac{\epsilon_{h,rup}}{\nu_{secu} \epsilon_{co}} + \frac{(1 - \nu_{secu} - 2\nu_{secu}^2)}{\nu_{secu}} \left(\frac{E_{frp} t}{E_{seco} R} \right) \\ & \times \left(\frac{\epsilon_{h,rup}}{\epsilon_{co}} \right) + \frac{4(k_1 - 1)(1 - \nu_{secu} - 2\nu_{secu}^2)}{\sqrt{3} \nu_{secu}} \\ & \times \left(\frac{E_{frp} t}{E_{seco} R} \right)^2 \left(\frac{\epsilon_{h,rup}}{\epsilon_{co}} \right)^2 \end{aligned} \quad (18)$$

where ν_{secu} = secant Poisson's ratio of the confined concrete under the ultimate condition and the term $E_{frp} t / E_{seco} R$ is the confinement stiffness ratio, representing the stiffness ratio between the FRP jacket and the concrete core.

The definitions of the secant moduli E_{sec} and E_{secu} are given in Fig. 10, which can be found by subtracting Eq. (11b) from Eq. (11a). The secant Poisson's ratio can be found from the constitutive equations (Eqs. (11a), (11b) and (11c)) and related to the lateral-to-axial strain ratio ϵ_r / ϵ_c through the following equation:

$$\nu_{sec} = \frac{\frac{\sigma_r}{\sigma_c} - \frac{\epsilon_r}{\epsilon_c}}{1 - 2 \frac{\epsilon_r}{\epsilon_c} \frac{\sigma_r}{\sigma_c} + \frac{\sigma_r}{\sigma_c}} \quad (19)$$

6.2. Proposed equation

Eq. (18) shows clearly the dependence of the ultimate strain of FRP-confined concrete ϵ_{cu} on the confinement stiffness ratio and the strain capacity of FRP at hoop rupture. Existing test data on FRP-confined concrete also show that the secant Poisson's ratio of FRP-confined concrete under the ultimate condition ν_{secu} depends strongly and predominantly on the confinement stiffness ratio (Fig. 11). Thus, the ultimate strain of FRP-confined concrete ϵ_{cu} can be taken to be a function only of the confinement stiffness ratio $E_{frp} t / (E_{seco} R)$ and the strain ratio $\epsilon_{h,rup} / \epsilon_{co}$. The following general equation is, therefore, proposed to predict the ultimate strain:

$$\epsilon_{cu} / \epsilon_{co} = c + k_2 \left(\frac{E_{frp} t}{E_{seco} R} \right)^\alpha \left(\frac{\epsilon_{h,rup}}{\epsilon_{co}} \right)^\beta \quad (20)$$

where c = normalized ultimate strain of unconfined concrete, k_2 = strain enhancement coefficient, and α and β are exponents to be determined. The effect of the secant Poisson's ratio ν_{secu} is reflected by the choice of appropriate values for α , β and k_2 .

Eq. (20) explicitly accounts for the stiffness and the actual ultimate condition of the jacket. If both α and β are taken as unity, Eq. (20) reduces to

$$\epsilon_{cu} / \epsilon_{co} = c + k_2 \frac{f_{l,a}}{f'_{co}} \quad (21)$$

which then relates the normalized ultimate strain to the actual confinement ratio only.

6.3. Determination of α and β

The values of the two exponents, α and β , are determined here using the test data of the present

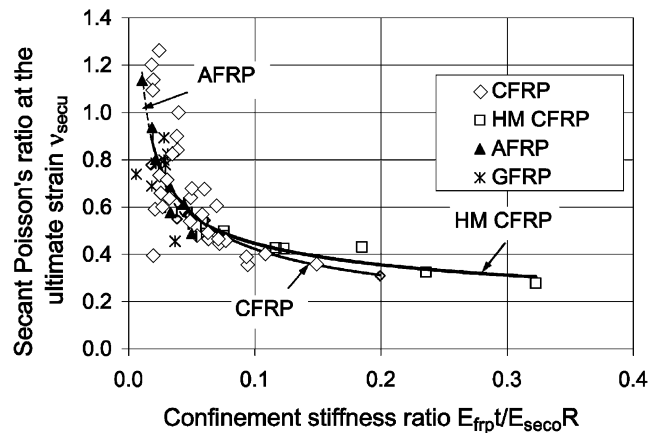


Fig. 11. Dependence of ultimate secant Poisson's ratio on confinement stiffness ratio.

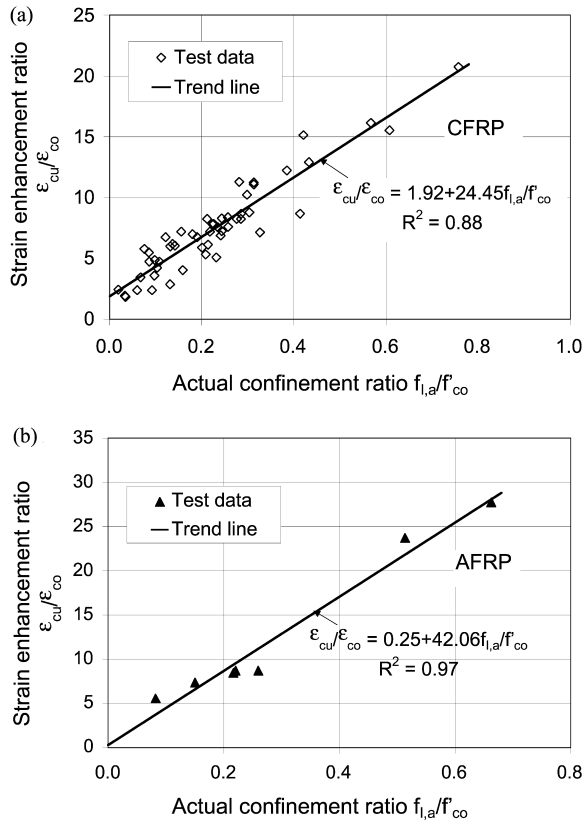


Fig. 12. Strain enhancement ratio vs. actual confinement ratio. (a) CFRP wraps; (b) AFRP wraps.

database. In determining these exponents, the strain at the compressive strength of unconfined concrete ε_{co} was taken as 0.002, a value, which was assumed in all data interpretation in the present study.

Fig. 12 shows two plots of the strain enhancement ratio against the actual confinement ratio from the tests for CFRP wraps and AFRP wraps, respectively. A linear relationship clearly exists in both cases but the two trend lines are very different. These diagrams indicate that the ultimate strain of FRP-confined concrete can be related linearly to the actual confinement ratio for a given type of FRP, but separate expressions are needed for different types of FRP due to differences in stiffness. To achieve a unified expression for the ultimate strain of FRP-confined concrete, the confinement stiffness ratio needs to be included, which means that the exponents, α and β , cannot both be unity. Thus, the following expression is suggested for FRP-wrapped concrete based the trends of the test data:

$$\begin{aligned} \varepsilon_{cu}/\varepsilon_{co} &= 1.75 + 12 \left(\frac{E_{frp}t}{E_{seco}R} \right) \left(\frac{\varepsilon_{h,wrap}}{\varepsilon_{co}} \right)^{1.45} \\ &= 1.75 + 12 \left(\frac{f_{la}}{f'_{co}} \right) \left(\frac{\varepsilon_{h,wrap}}{\varepsilon_{co}} \right)^{0.45} \end{aligned} \quad (22)$$

Fig. 13a,b show that the trends of the test data are similar for both CFRP and AFRP wraps. For HM CFRP-wrapped specimens (Fig. 13c), if one of the three specimens tested by Dias da Silva and Santos (2001) is excluded (shaded) as a statistical outlier, the remaining specimens also show a trend similar to those observed for CFRP or AFRP wraps. A large scatter is observed for GFRP-wrapped specimens (Fig. 13d), the cause of which is difficult to pinpoint at the present, but the predictions of Eq. (22) are well covered by the scatter of the test data. When all test data are plotted together (Fig. 13e), a close overall agreement between the test data and Eq. (22) is observed. Eq. (22), therefore, provides a unified expression for the ultimate strain of FRP-confined concrete that is applicable to different types of FRP. Obviously, more test data for GFRP wraps should be obtained in the future for further verification of Eq. (22).

7. Compressive strength of confined concrete

The compressive strength of FRP-confined concrete has been discussed in detail in Lam and Teng (2002b). In that paper, the compressive strength of FRP-confined concrete f'_{cc} is related to the nominal confinement ratio through

$$\frac{f'_{cc}}{f'_{co}} = 1 + 2 \frac{f_l}{f'_{co}} \quad (23)$$

Fig. 14 is a plot of the strengthening ratio f'_{cc}/f'_{co} against the actual confinement ratio f_{la}/f'_{co} of the present test data. The trend line of these test data can be closely approximated using the following equation:

$$\frac{f'_{cc}}{f'_{co}} = 1 + 3.3 \frac{f_{la}}{f'_{co}} \quad (24)$$

This equation implies that in terms of the actual confinement ratio, the confinement effectiveness coefficient k_1 becomes 3.3 instead of 2. It should be noted that the stiffness of the confining jacket also has an effect on the compressive strength of concrete with the same confinement ratio, as demonstrated in Fig. 9. As this effect on the compressive strength is far less than that on the ultimate strain, it is ignored in Eq. (24).

It should be reminded that a significant strength enhancement can only be expected with an actual confinement ratio $f_{la}/f'_{co} \geq 0.07$. The use of Eq. (24) is recommended to be subjected to this condition. For the case of FRP-confined concrete with $f_{la}/f'_{co} < 0.07$, no strength enhancement is assumed.

8. Intercept of the stress axis by the linear second portion

The intercept of the stress axis by the linear second portion, f_o , is an important parameter in the proposed

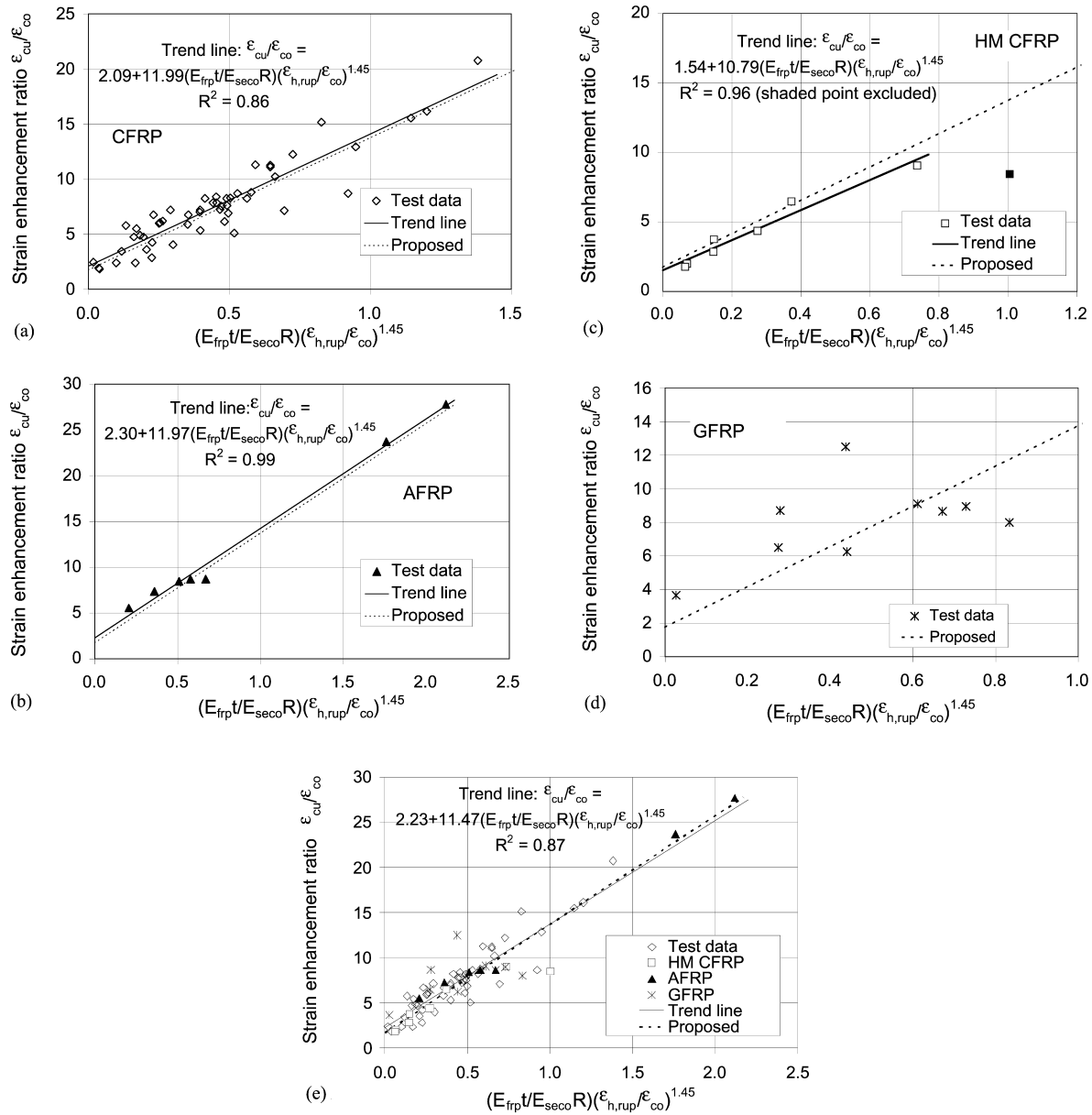


Fig. 13. Performance of proposed equation for the ultimate strain of FRP-confined concrete. (a) CFRP wraps; (b) AFRP wraps; (c) HM CFRP wraps; (d) GFRP wraps; (e) all specimens.

stress–strain model as it, together with the ultimate point, determines the slope of the second portion. Samaan et al. [6] proposed the following expression of f_o based on experimental data available to him:

$$f_o = 0.872f'_{co} + 0.371f_l + 6.258 \quad (\text{MPa}) \quad (25)$$

In the present database, values of f_o obtained by the present authors from test stress–strain curves have been included. These values are normalized by the compressive strength of unconfined concrete f'_{co} and are plotted against the actual confinement ratio $f_{l,a}/f'_{co}$ as shown in Fig. 15. The values of f_o/f'_{co} are seen to fall between

1.0 and 1.2 in most cases, which appear to be independent of the confinement ratio. For the 63 specimens, for which the values of the intercept are available, the average ratio of f_o/f'_{co} is 1.09 with a standard deviation of 0.13. It is, therefore, suggested for simplicity that in the proposed model

$$f_o = f'_{co} \quad (26)$$

It should be noted that with $f_o = f'_{co}$, the second linear portion of the proposed stress–strain model reduces to a horizontal straight line (i.e. $E_2 = 0$) as assumed in both BS 8110 [40] and Eurocode 2 [15], and at the same

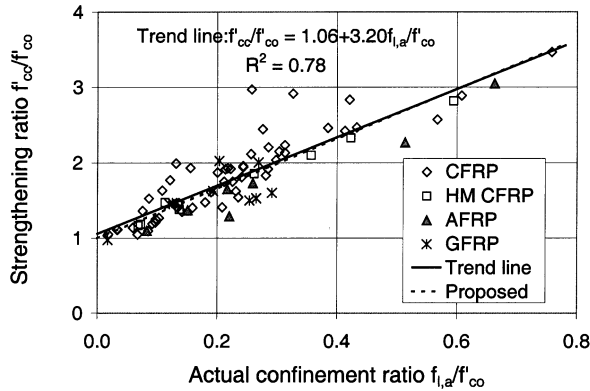


Fig. 14. Strengthening ratio vs. actual confinement ratio.

time, Eq. (8a) reduces to Hognestad's parabola for unconfined concrete. That is, the present model for FRP-confined concrete reduces directly to the design stress–strain models in BS 8110 [40] and Eurocode 2 [15] for unconfined concrete, provided the same initial elastic modulus is used.

9. FRP efficiency factor

In the preceding sections, the definitions of the ultimate strain, compression strength and minimum amount of FRP for sufficient confinement for the proposed stress–strain model are all in terms of the actual confinement ratio, so the actual hoop rupture strain of the FRP is required. To facilitate the application of the proposed stress–strain model, an FRP efficiency factor is, therefore, proposed here, which is defined as the ratio of the actual FRP hoop rupture strain ($\varepsilon_{h,rupt}$) in FRP-confined concrete to the FRP rupture strain from flat coupon tests (ε_{frp}). For the 52 CFRP-wrapped specimens, out of the total of 76 specimens in the present database (Table 1), this efficiency factor is 0.586 on average. Making use of this efficiency factor, the ultimate strain of CFRP-confined concrete can be expressed as

$$\varepsilon_{cu}/\varepsilon_{co} = 1.75 + 5.53 \left(\frac{f_{l,a}}{f'_{co}} \right) \left(\frac{\varepsilon_{frp}}{\varepsilon_{co}} \right)^{0.45} \quad (27)$$

With this equation, the user only needs to know the tensile strain ε_{frp} from flat coupon tests. This average efficiency factor of 0.586 for CFRP-wrapped concrete also unifies Eq. (23) and Eq. (24).

For other types of FRP, insufficient information exists to define this efficiency factor with confidence. Even for CFRP-confined concrete, there is a considerable scatter in the efficiency factor deduced from test results. The present authors, therefore, recommend that for cost-effective and safe applications, a small number (say 3) of FRP-confined concrete cylinder tests, analogous to

plain cylinder tests, be conducted to determine the efficiency factor of a given FRP product in confinement applications. For this purpose, a standard confined cylinder test method should be formulated in the future. FRP manufacturers can then supply the results of such tests as part of their product information. If this information is not available from the manufacturer, the user should conduct these tests instead.

10. Comparison with test data

The proposed stress–strain model for FRP-confined concrete is compared with the test data obtained by Xiao and Wu [11] on CFRP-wrapped concrete cylinders, as shown in Fig. 16. Details of these specimens can be found in Table 1a. Fig. 16a and b are for specimens having an unconfined concrete strength f'_{co} of 55.2 MPa and wrapped with one (specimens 41–43) and two (specimens 44 and 45) layers of CFRP, respectively. Fig. 16c is for specimens having $f'_{co}=43.8$ MPa and wrapped with three layers of CFRP (specimens 38–40), and Fig. 16d is for specimens having $f'_{co}=33.7$ MPa and wrapped with three layers of CFRP (specimens 30 and 31). Each figure shows test stress–strain curves from nominally identical specimens and predictions of the present model. For each case, two predicted curves are shown: one with the FRP hoop rupture strain $\varepsilon_{h,rupt}$ estimated on the basis of the average efficiency factor for CFRP of 0.586 wrapped specimens (Table 2) and the other with the hoop rupture strain being the average of the actual values recorded during the tests. The elastic modulus of unconfined concrete was taken as $E_c=4730\sqrt{f'_{co}}$ [48] and the strain at the compressive strength of unconfined concrete was taken as $\varepsilon_{co}=0.002$ for calculating the ultimate strain from Eq. (22). The specimens used in Fig. 16a had an average actual confinement ratio of $f_{l,a}/f'_{co}=0.048$, so these specimens

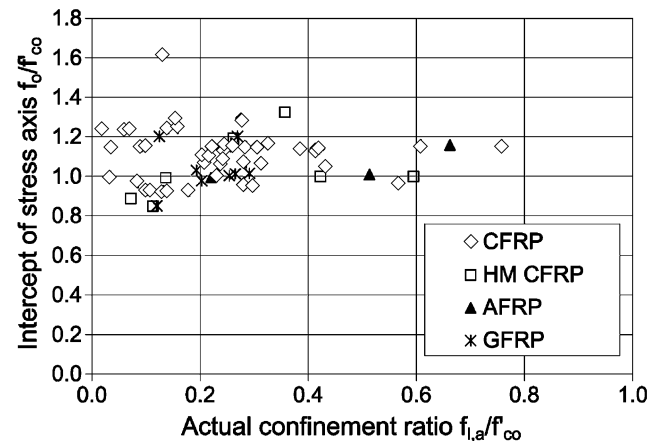


Fig. 15. Intercept of stress axis by the second linear portion vs. actual confinement ratio.

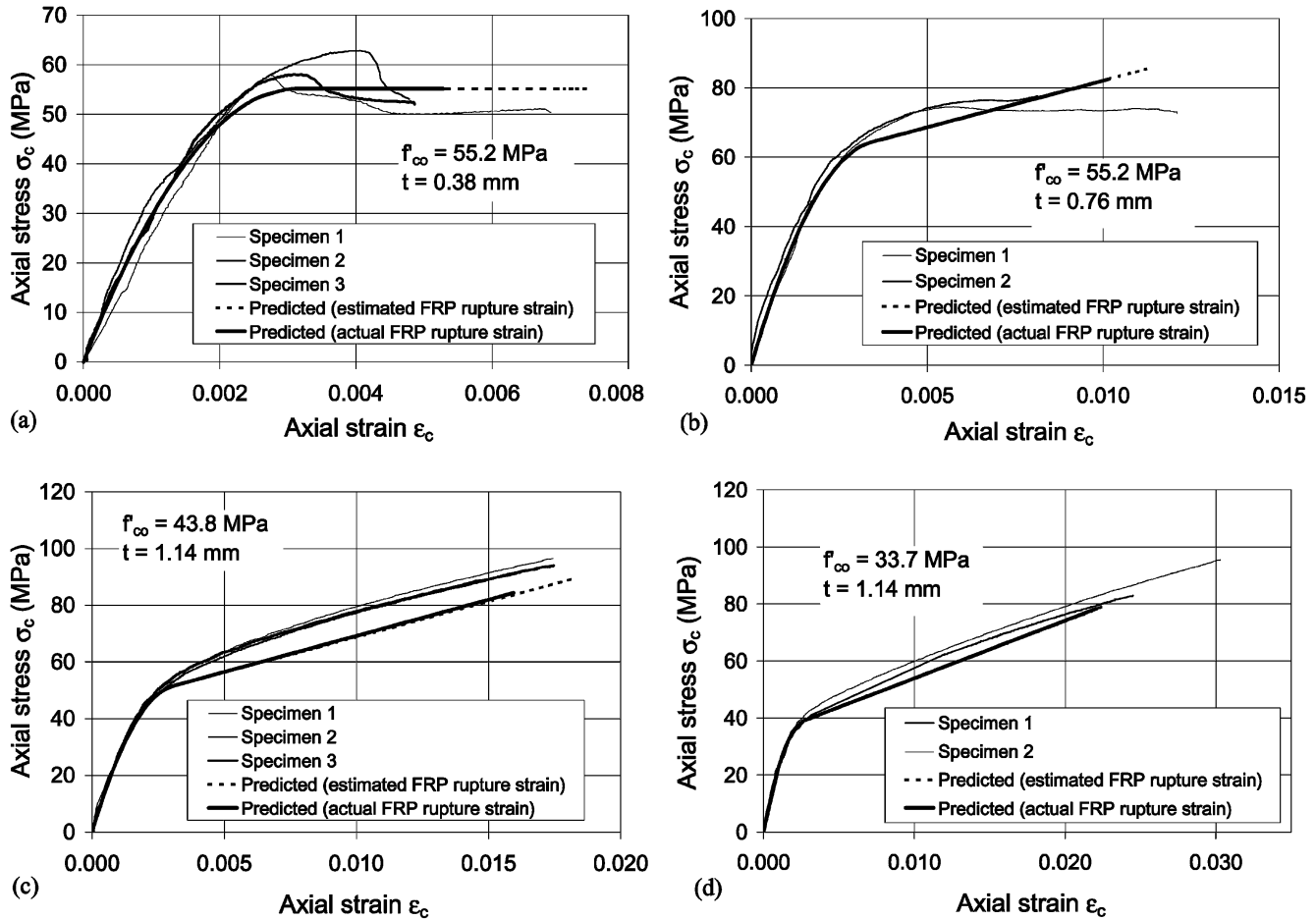


Fig. 16. Comparison between proposed model and test stress-strain curves. (a) $f_{t,a}/f'_{co} = 0.048$; (b) $f_{t,a}/f'_{co} = 0.150$; (c) $f_{t,a}/f'_{co} = 0.281$; (d) $f_{t,a}/f'_{co} = 0.403$.

are insufficiently confined ones. Consequently, $k_1 = 0$ was used in predicting the compressive strength of confined concrete. The average actual confinement ratios for the specimens included in Fig. 16b,c and d are 0.150, 0.281 and 0.403, respectively. For these cases, $k_1 = 3.3$ was used. It can be seen from Fig. 16 that the predictions compare well with the test results. It should be noted that the ultimate strains of specimens 41–43 are all considerably overestimated using the estimated $\epsilon_{h,rupt}$ (Fig. 16a), which further justifies the exclusion of such insufficiently confined concrete from practical considerations and from the range of applicability of the proposed stress-strain model.

11. Conclusions

This paper has been concerned with the development of a stress-strain model for concrete confined by wrapped FRP with fibres only or predominantly in the hoop direction. Existing experimental data have been thoroughly reviewed and discussed, and the deficiencies of existing stress-strain models are highlighted. A sim-

ple and accurate stress-strain model for FRP-confined concrete has been presented for design use. The results and discussions presented in this paper also allow the following conclusions to be drawn:

1. The average hoop strain in FRP at rupture in FRP-wrapped concrete can be much lower than the FRP material ultimate tensile strain from flat coupon tests, indicating the assumption that FRP ruptures when the FRP material tensile strength reached is not valid in the case of concrete confined by wrapped FRP. Based on this observation, a unified stress-strain model for concrete confined by different types of FRP must be based on the actual hoop rupture strain of FRP rather than the ultimate material tensile strain.
2. The stress-strain curve of FRP-confined concrete can be in one of several forms, but in the vast majority of cases, this curve is or can be approximated as a monotonically ascending bi-linear curve. Such FRP-confined concrete is said to be sufficiently confined. Any FRP-confined concrete with an actual confinement ratio less than 0.07 is said to be insufficiently-

confined. Such concrete is not expected to possess a compressive strength significantly above that of unconfined concrete and the FRP may rupture at a low hoop strain. Such insufficiently-confined concrete should not be allowed in design.

3. The new design-oriented stress–strain model proposed in this paper is in a form that is familiar to engineers. This model is simple, so it is suitable for direct use in design, but in the meantime, it captures all the main characteristics of the stress–strain behavior of FRP-confined concrete. The model reduces directly to idealized stress–strain curves adopted by existing design codes for unconfined concrete.
4. The stiffness of the FRP jacket and the actual ultimate condition of the jacket are explicitly accounted for in the proposed model. As a result, the proposed model is applicable to concrete confined by different types of FRP. The predictions of the model have been shown to agree well with a set of test data.
5. For the application of the proposed model in design, the FRP efficiency factor (ratio between the actual hoop rupture strain of FRP in FRP-confined concrete and the ultimate tensile strain from material tests) needs to be established. For this purpose, confined cylinder tests have been suggested as a standard type of test to supply this information.

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